Physics Questions

Vectors and Kinematics

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A research rover is located at the origin of an xy-plane. A sensor beacon is at position $\vec{r} = -9\hat{i} + 12\hat{j}$, where distances are measured in kilometers. Which statement is correct?

A. The beacon is 21 km from the rover.
This adds the magnitudes of the components, $9 + 12$, instead of calculating the magnitude of the vector. Distance from the origin must be found using $|\vec{r}| = \sqrt{x^2 + y^2}$, not by simple addition.

B. The beacon is 12 km from the rover.
This uses only the y-component as the full distance. In two dimensions, both the x- and y-components contribute to the magnitude.

C. The beacon is 15 km from the rover.
The distance from the origin is the magnitude of the position vector. Using $|\vec{r}| = \sqrt{(-9)^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15\,\text{km}$, the beacon is 15 km away. This applies the Pythagorean theorem to the perpendicular vector components.

D. The beacon is 9 km from the rover.
This uses only the x-component's magnitude as the total distance. A single component is not the same as the full vector magnitude unless the other component is zero.

Answer: *C*

Q2. The position of a cart moving along the x-axis is given by $\vec{s}(t) = \left[(2.4\,\text{m/s}^3)t^3 - (1.1\,\text{m/s}^2)t^2\right]\hat{i}$. Which statement is correct?

A. This motion has constant acceleration.
A cubic position function does not lead to constant acceleration. Taking two derivatives gives an acceleration that still depends on t.

B. The magnitude of the acceleration at $t = 0$ is $2.2\,\text{m/s}^2$.
Differentiate the position twice: $v(t) = \frac{ds}{dt} = 7.2t^2 - 2.2t$ and $a(t) = \frac{dv}{dt} = 14.4t - 2.2$. At $t = 0$, the acceleration is $a(0) = -2.2\,\text{m/s}^2$, so its magnitude is $2.2\,\text{m/s}^2$. The correct statement focuses on the magnitude, not the sign.

C. The acceleration is always $1.1\,\text{m/s}^2$.
This incorrectly treats the coefficient of the t^2 term in position as the acceleration itself. Acceleration comes from the second derivative of position, not directly from a coefficient in $s(t)$.

D. The magnitude of the acceleration at $t = 0$ is $0\,\text{m/s}^2$.
This mistake comes from assuming every term vanishes at $t = 0$ after differentiation. The second derivative leaves the constant term $-2.2\,\text{m/s}^2$, so the magnitude is not zero.

Answer: *B*

Q3. A particle takes 4.0 s to move from position $\vec{r}_A = 3\hat{i} - 2\hat{j}$ m to position $\vec{r}_B = -9\hat{i} + 7\hat{j}$ m. Which statement is correct?

A. The distance traveled by the particle is 15 m.
The value 15 m is the magnitude of the displacement, not necessarily the actual path length. Distance traveled depends on the path taken, which is not specified.

B. The speed at $t = 0$ must be 3.75 m/s.
This divides displacement magnitude by elapsed time to get an average rate and then incorrectly treats it as the initial speed. Instantaneous speed at $t = 0$ cannot be determined from only the starting point, ending point, and total time.

C. The acceleration at $t = 0$ must be 0.94 m/s^2.
Acceleration requires more detailed information about how velocity changes with time. Initial and final position plus total time are not enough to determine the acceleration at a specific instant.

D. None of the above is correct.
From the given data, the displacement is $\Delta\vec{r} = \vec{r}_B - \vec{r}_A = (-12\hat{i} + 9\hat{j})$ m, whose magnitude is $|\Delta\vec{r}| = \sqrt{(-12)^2 + 9^2} = 15$ m. However, that value is not guaranteed to equal the distance traveled, and the initial speed and initial acceleration cannot be inferred from the limited information provided. Therefore A, B, and C are all false.

Answer: *D*